Prior Design Attempt #1 Sammi

Romans Cement Pipes 2000 BC

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The Romans' transportation of water began around 2000 BC after they invented the cement pipe. The pipes helped transport water from the source to designated areas. The pipes ran underground to provide other destinations with water. The cement pipes were an early start to the irrigation system through their idea of transporting water. The idea of transporting water was a gateway for the irrigation system to rise and become well known and highly used. The pipes were very successful and helped grow the improvement of water transportation and irrigation systems.

Noria Irrigation System | Design 2

Peter Schwindt

The Noria, also known as a water wheel, is believed to have been invented in the 3rd century B.C. They originated in Asia but made their way to Europe and were very popular there. The Noria moves water from a lower elevation to a higher elevation. They are strategically placed on running rivers in order for the water flow to spin the wheel while simultaneously filling the buckets. The buckets are then dumped into a wooden half pipe which uses gravity to distribute the water to the desired location.

Photo Credit: Machinery Lubrication

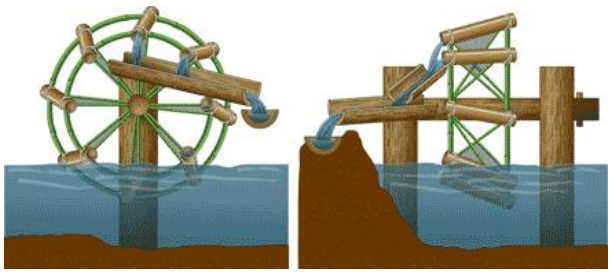


Photo Credit: Tripadvisor

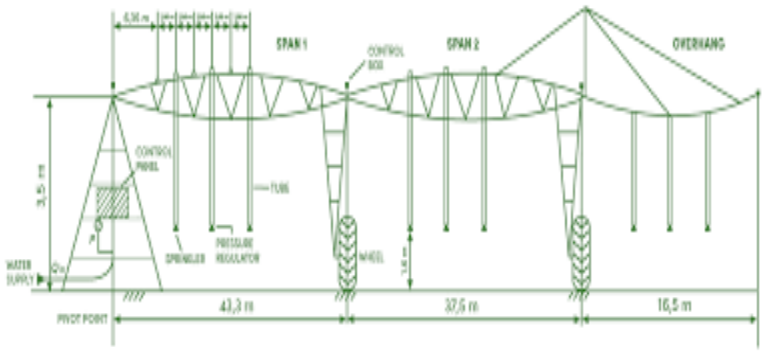
Photo Credit: Kuriositas

Source:

https://en.wikipedia.org/wiki/Noria#:~:text=Its%20concept%20is%20similar%20to,than%20by%20animals%20or%20motors.

Center Pivot Irrigation System | Design 3 Aidan Chavez

In 1948, Nebraska farmer Frank Zybach invented the first center-pivot irrigation system after watching farmers laboriously move ground irrigation pipes feeding sprinkler heads across farm fields. The original and most used style of pivot irrigation applies water in a circular pattern, pivoting around a central point in the middle of the field.



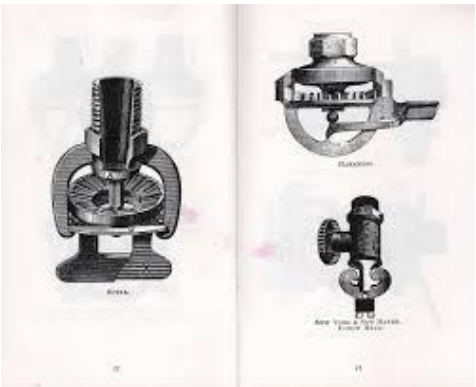
Cite-https://www.agrivi.com/blog/center-pivot-system-an-efficient-and-economical-solution-for-irrigation/#:~:text=In%201948%2C%20Nebraskan%20farmer%20Frank,sprinkler%20heads%20across%20farm%20fields.

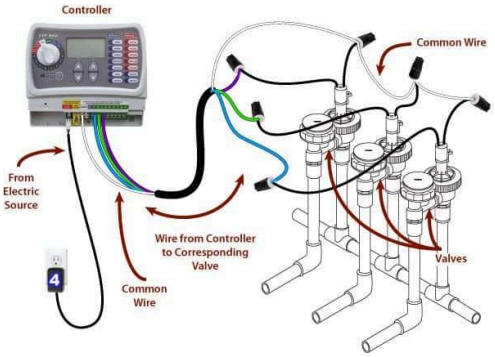
Prior Design Attempt #1 Lauren

Sprinkler Control System

The original sprinkler system was a large pipe connected to a smaller, perforated pipe which would be attached all around the building.

The modern controller used today can do much more. It activates and deactivates the sprinkler, controls how much space is covered, allocates the correct amount of water, and activates all the mechanisms for consistent operations.





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Drip irrigation system | design 5

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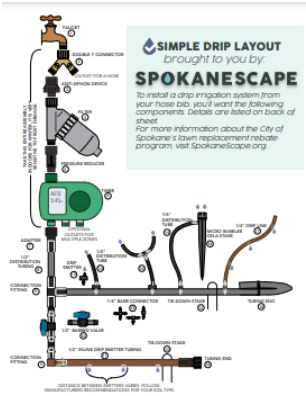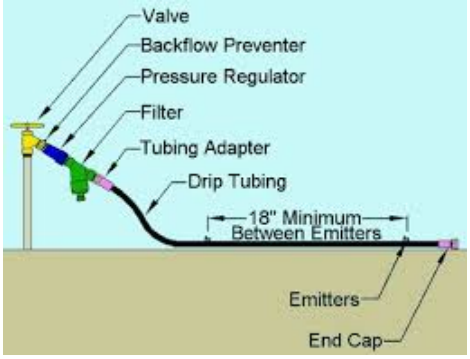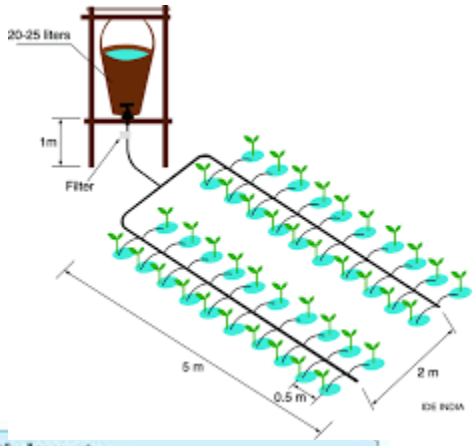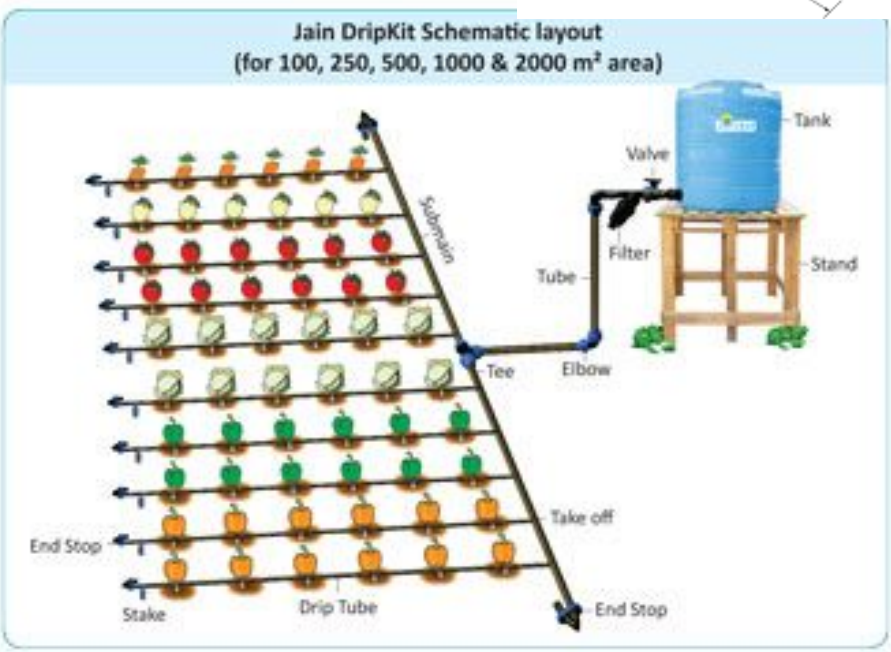
Modern drip irrigation was invented in Israel in the 1960s, such irrigation also improved the efficiency of fertilizer use as well as water. The introduction of this technology was revolutionary according to the Times of Israel and could save 70% of water usage in the growth of rice.